

Exp- 23 Forward Chaining

Program:

```
facts = ["fever", "cough"]

rules = {
    "fever and cough": "flu",
    "flu": "take rest"
}

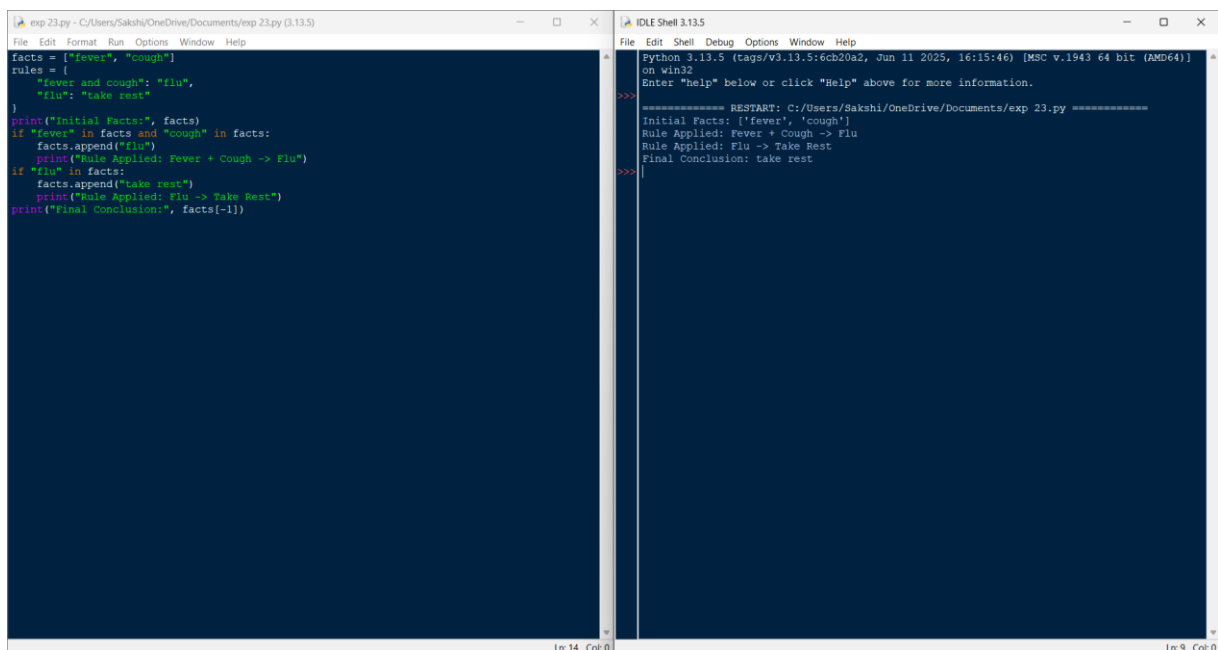
print("Initial Facts:", facts)

if "fever" in facts and "cough" in facts:
    facts.append("flu")
    print("Rule Applied: Fever + Cough -> Flu")

if "flu" in facts:
    facts.append("take rest")
    print("Rule Applied: Flu -> Take Rest")

print("Final Conclusion:", facts[-1])
```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'exp 23.py', contains the following code:

```
facts = ["fever", "cough"]
rules = {
    "fever and cough": "flu",
    "flu": "take rest"
}
print("Initial Facts:", facts)
if "fever" in facts and "cough" in facts:
    facts.append("flu")
    print("Rule Applied: Fever + Cough -> Flu")
if "flu" in facts:
    facts.append("take rest")
    print("Rule Applied: Flu -> Take Rest")
print("Final Conclusion:", facts[-1])
```

The right window, titled 'IDLE Shell 3.13.5', shows the output of the program:

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sakshi/OneDrive/Documents/exp 23.py =====
Initial Facts: ['fever', 'cough']
Rule Applied: Fever + Cough -> Flu
Rule Applied: Flu -> Take Rest
Final Conclusion: take rest
>>>
```